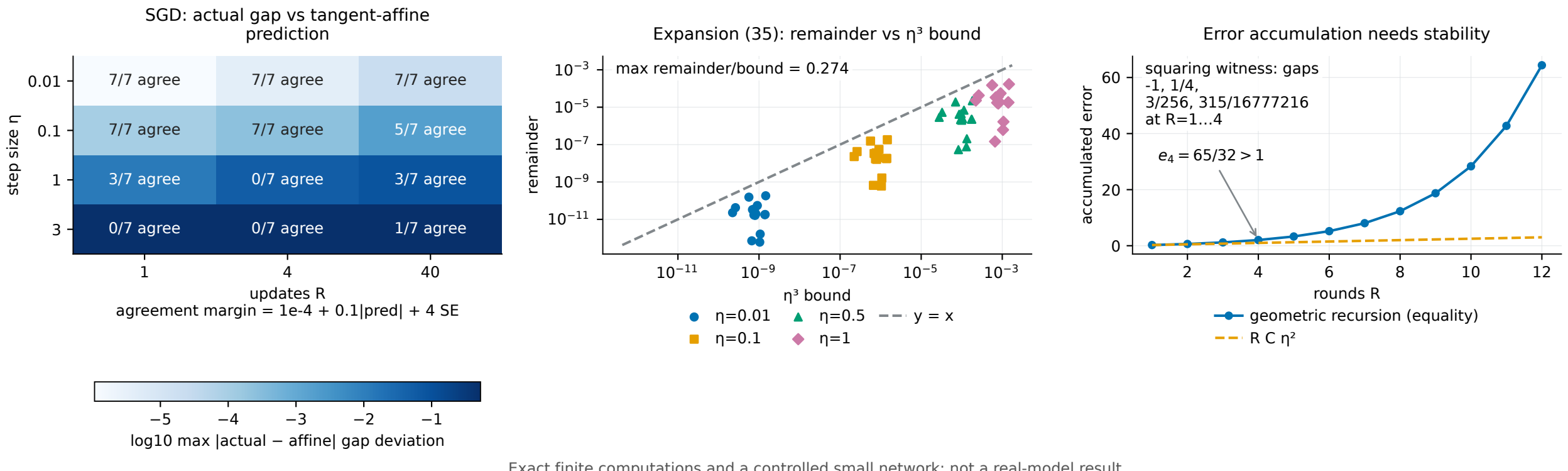
SGD: actual gap vs tangent-affine prediction
step size η
0.01
0.1
1
3
7/7 agree
7/7 agree
7/7 agree
7/7 agree
7/7 agree
5/7 agree
3/7 agree
0/7 agree
3/7 agree
0/7 agree
0/7 agree
1/7 agree
1
4
40
updates R
agreement margin = 1e-4 + 0.1|pred| + 4 SE
−5
−4
−3
−2
−1
log10 max |actual − affine| gap deviation
Expansion (35): remainder vs η³ bound
max remainder/bound = 0.274
remainder
η³ bound
η=0.01
η=0.5
y = x
η=0.1
η=1
Error accumulation needs stability
squaring witness: gaps
-1, 1/4,
3/256, 315/16777216
at R=1…4
$e_4 = 65/32 > 1$
accumulated error
rounds R
geometric recursion (equality)
R C η²
Exact finite computations and a controlled small network; not a real-model result.